1st year | 2024-2025 | Lecture 10

Quantifying Risk

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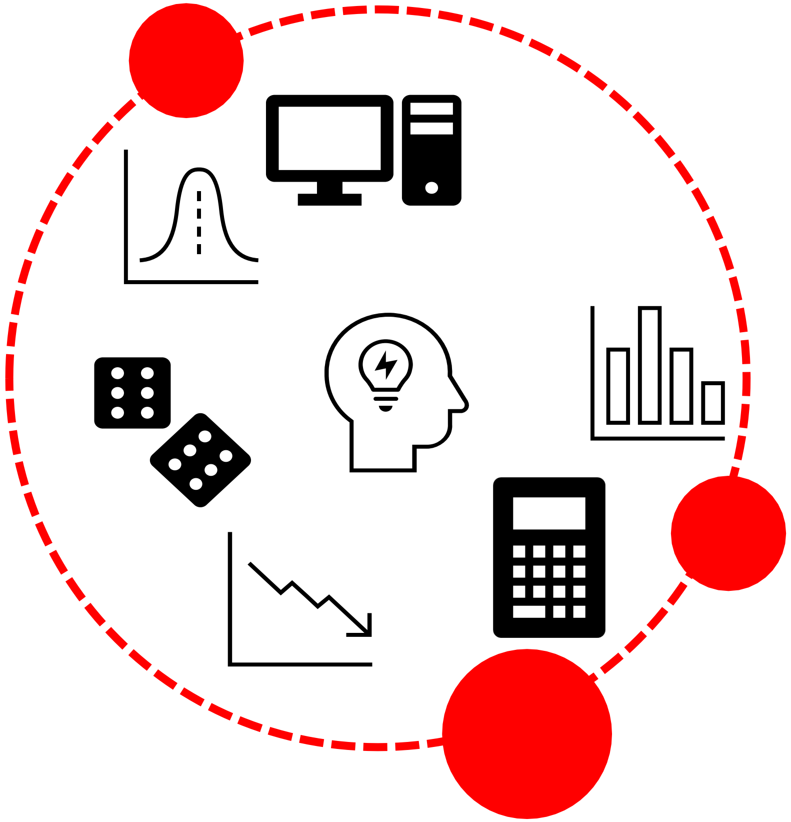

General points

- You can use the chi-square test to determine if there is a statistically significant association between a disease and a factor. On the other hand, Relative Risk (RR) and Odds Ratio (OR) are used to quantify the strength of this association, measuring how strongly the factor is related to the disease.

		Disease or condition		
		+ve	- ve	total
Factor	Group 1 (exposed)	a	b	a + b
	Group 2 (non-exposed)	c	d	c + d
		a + c	b + d	N

Relative Risk (RR)

- Relative Risk (RR), also known as the risk ratio, compares the probability of a particular condition (such as a disease) occurring in two different groups. It is commonly used in cohort studies.

$$RR = \frac{\text{Disease risk (incidence) in exposed group}}{\text{Disease risk (incidence) in non-exposed group}}$$

$$RR = \frac{a}{a+b} / \frac{c}{c+d}$$

Interpretation of Results

- RR = 1 means that exposure does not affect the outcome (disease or condition). There is no association between the exposure and the condition.

- RR > 1 means that the risk of the disease is increased by the exposure indicating a positive association.

- Example: If RR = 1.5, animals with exposure have a 50% higher risk of developing the disease compared to those without the exposure.

Interpretation of Results

- RR < 1 means that the risk of the disease is decreased by the exposure indicating a negative association or protective effect.

- Example : If RR = 0.5, this means that animals in the exposed group only have half the risk of developing the disease compared to animals in the non-exposed group. The condition is protective.

Example 1

A veterinary researcher is conducting a study to investigate the association between a specific dietary factor (Diet X) and the development of a gastrointestinal disorder (GID) in cats. The researcher identifies two groups of cats: Group X, which has been exposed to Diet X, and Group Y, which has not been exposed to Diet X. After a follow-up period, the researcher records the occurrence of GID in each group as shown in the table. Herein GID is the disease, and diet is the risk factor.

	GID +ve	GID -ve	Total	Risk
Diet X	33	117	150	0.22
Diet Y	15	135	150	0.10
Total	45	255	300	RR = 0.22/0.10 = 2.20

[SPSS, chi-square = 8.036, P = 0.005; OR = 2.538 ; RR = 2.20]

Interpretation:

- ✓ RR = 2.2 indicates that cats exposed to Diet X have 2.2 times the risk of developing GID compared to cats not exposed to Diet X.

- ✓ This reflects a 120% increase in risk, calculated as (2.2−1)×100% = 120%.

- ✓ In summary, the RR of 2.2 suggests a positive association between Diet X and GID, meaning that Diet X is associated with an increased likelihood of developing GID.

Attributable Risk (Risk Difference)

- The risk difference between the two groups (risk in group 1 - risk in group 2) is termed the attributable risk (AR).

- It is used to answer the question: "What proportion of the disease is due to exposure to the risk factor?" In the above example, What proportion of GID is due to feeding Diet X?

- Risk of GID among cats on Diet X = 0.22 Risk of GID among cats on Diet Y = 0.10 AR = 0.22 - 0.10 = 0.12 (12%)

- AR of 12% means that 12% of the cases of GID among cats exposed to Diet X can be attributed to the consumption of Diet X.

- In other words, if Diet X were removed or not consumed, 12% of the GID cases in the group exposed to Diet X would not have occurred.

Odds Ratio (OR)

- **Odds :** The odds are the probability that something (e.g., disease) will happen divided by the probability that it will not happen.

$$\text{Odds} = \frac{P}{1 - P}$$

- If you flip a coin, the probability of getting a head is 0.5. The odds you will get a head are 1 (0.5/1-0.5).

- **Odds Ratio (OR)** : is the ratio of the odds in the exposed group to the odds in the unexposed group.

  The OR is commonly used in case-control and cross-sectional studies.

$$\text{OR} = \frac{\text{Odds of Outcome in Exposed Group}}{\text{Odds of Outcome in Unexposed Group}}$$

Odds Ratio (OR)

Calculating the Odds Ratio in 2 x 2 tables

Refer to the basic 2 x 2 table given above,

$$\text{Group 1, Odds} = \frac{a}{a+b} / \frac{b}{a+b} = \frac{a}{b}$$

$$\text{Group 2, Odds} = \frac{c}{c+d} / \frac{d}{c+d} = \frac{c}{d}$$

$$OR = \frac{a}{b} / \frac{c}{d} = \frac{ad}{bc}$$

Example 2

- To examine the relationship between lameness and repeat breeding in dairy cows, we conducted a case- control study. We compared 150 cows that experienced repeat breeding (requiring more than two inseminations to conceive) with 200 cows that conceived after one or two inseminations, focusing on the presence of lameness before conception.

	Lame	Not Lame	Total
Case (Repeat breeders)	50	100	150
Control (Non-Repeat breeders)	20	180	200
Total	70	280	350

[SPSS, chi-square = 29.17, P = < 0.001; OR = 4.5; RR = 3.3]

In this example, the odds ratio of 4.5 suggests that cows with lameness are 4.5 times more likely to be repeat breeders compared to cows without lameness.

Example 3

- A group of 3000 adult dairy cows was classified as whether or not they had a particular disease shortly after calving. Based on the previous dry period length, the cows were also classified as having received a recommended (45-70 days) or short (< 45 days) length of dry period.

	Disease (+ve)	Disease (-ve)	Total
Short dry period	20	1250	1270
Recommended dry period	10	1720	1730
Total	30	2970	3000

$$RR = \frac{20/1270}{10/1730} = \frac{0.01575}{0.00578} = 2.72$$

$$OR = \frac{20 \text{ x } 1720}{1250 \text{ x } 10} = \frac{34400}{12500} = 2.75$$

- [SPSS, OR = 2.752; RR = 2.724]

- Notice that, OR in the previous example is nearly similar to RR because the disease is rare

Thank you

M.Marie